

The goal of the article is to highlight and summarize your work & learning from this summer. The audience is ACEP's broader community, professionals across the global energy research sector, and community members in Alaska.

This article is valuable for you as a young professional, and to showcase the program and for funding support. The article will be featured on ACEP's news feed and posted to ACEP's LinkedIn and Facebook pages.

Helping Secure Rural Alaska's Microgrids

As energy infrastructure becomes increasingly digitalized, its cybersecurity and networking systems must keep pace. In rural Alaska however, communities operating local microgrids often lack access to the necessary technical expertise and training resources to maintain secure network infrastructure.

That's why Caden Kelly, a freshman electrical engineering and computer sciences student from UC Berkeley, is working as a summer intern with ACEP to create deployable Cyber Security training materials for rural Alaskan communities.

Kelly worked with mentor John Haverlack, ACEP's Information Security Lead. His project focused on creating a portable network training ground and two network training modules. The portable training ground allowed participants to interact with a network and firewall, then configure a similar system themselves, following the hands-on training modules developed by Kelly.

Many Alaskan communities continue to rely on aging cyber infrastructure with limited network segmentation and firewall protection. Kelly's training materials empower participants to build future networks with security by design, while demonstrating a cost effective solution to these resource-constrained communities. The goal was to give communities the knowledge and tools needed to better protect the power systems they rely on every day.

In tandem with these goals is Kelly's own development: "I came into the summer with almost no experience in Linux or computer networking, so everything was new to me. By the end of the internship, I'd built and configured complex virtual networks, firewalls and software-defined infrastructure that I never would have imagined working with just a few months earlier. John was great! He was always there to answer questions, but more importantly, he taught me better ways to troubleshoot problems on my own and think through them systematically. Networking has gone from something completely unfamiliar to something I really enjoy because every problem feels like solving a puzzle that ultimately helps protect systems people rely on every day."

This internship is funded by the National Science Foundation through ACEP's National Science Foundation Research Experiences for Undergraduates program. View the final presentation for this project on ACEP's YouTube Channel. For more information on this project, please contact John Haverlack at jehaverlack@alaska.edu.

For Yuri: Link to youtube: <https://www.youtube.com/@amandabyrdalaskacenterfore6293>

Link to mentor's bio on ACEP: <https://www.uaf.edu/acep/about/our-team/john-haverlack.php>

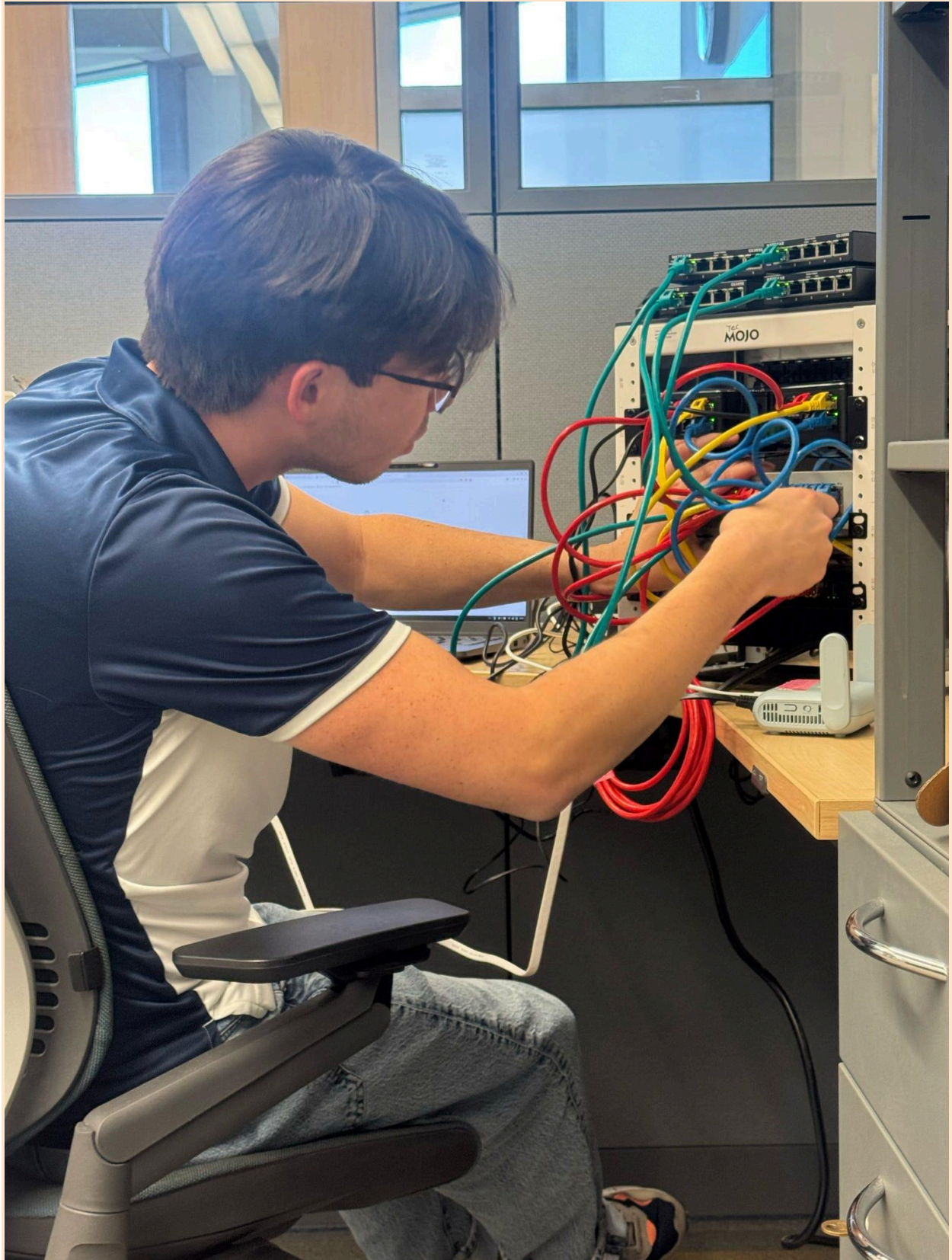
Photos (size ~1MB) and captions.

2026-Intern-LastName-description-PhotographerName

2026-Intern-Smith-data-analysis-PattyEagan

2026-Intern-Kelly-Network Building-Annabeth Icke

Description: Photo courtesy of other intern Annabeth Icke, while I am working on my network wiring.



2026-Intern-Kelly-Headshots-Yuri Bult-Ito

Description: Photo courtesy of Yuri, for professional photos day.

